

Exploring Excess Return Generation through European Fixed Income Arbitrages

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Duarte et al. (2007) pose the question that motivates this paper:

Do fixed-income arbitrage strategies earn genuine alpha, or are their returns simply compensation for systematic risk — “picking up nickels in front of a steamroller”?

Their answer: the more complex strategies generate **significant alpha** that survives a 10-factor benchmark in U.S. markets.

Open questions that motivate this paper:

- Does the result generalise to **European** fixed-income markets?
- Does it survive a **much more demanding** battery of factor controls?
- And — perhaps above all — **why does the alpha persist?**

RQ1. *Does alpha exist?*

Do three European fixed-income arbitrage strategies — the BTP Italia inflation-linked basis, the corporate CDS–Bond basis, and the iTraxx index skew — generate excess returns that survive standard benchmark factor models?

→ Section 3

RQ2. *Does alpha survive aggressive risk adjustment — and is it state-dependent?*

Does the alpha survive when the candidate factor universe is enlarged to **85** controls and selected by data-driven methods (rolling PCA; Adaptive Elastic Net with stability selection)? And: does the residual alpha vary systematically with intermediary funding stress?

→ Section 4

RQ3. *Why does alpha persist?*

Are the three mispricings linked to one another, and is the co-movement consistent with the slow-moving capital framework of Duffie (2010)?

→ Section 5

We extend Duarte et al. (2007) in four respects:

- 1 **New strategies, never studied in this combination.** To our knowledge, BTP Italia inflation-linked basis, European CDS–Bond basis in iTraxx Main, and iTraxx index skew across the four sub-indices have never been studied jointly as systematic arbitrages.
- 2 **Strategies with deterministic terminal payoff.** Unlike Duarte's swap-spread, yield-curve, volatility-selling and capital-structure trades, each of our three converges to a *certain* profit if held to maturity.
- 3 **Far more demanding factor controls.** Three classical benchmarks (Duarte et al. (2007), Brooks et al. (2020), Fung and Hsieh (2004)), *plus* rolling PCA on 85 candidate factors (Ludvigson and Ng, 2009), *plus* Adaptive Elastic Net (Zou and Zhang, 2009) with stability selection (Meinshausen and Bühlmann, 2010).
- 4 **We test *why* the alpha persists.** We map our findings to the predictions of the slow-moving capital literature (Duffie (2010), Brunnermeier and Pedersen (2009), Mitchell and Pulvino (2012), Boyarchenko et al. (2016)), and exploit holder-base heterogeneity in the European setting as the source of identification.

Basis definition (following Fleckenstein et al. (2014), adapted to the Italian instrument):

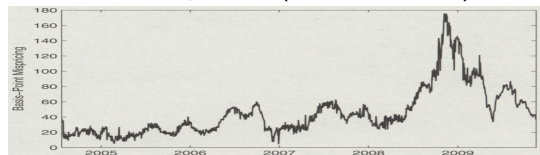
$$m_t^{\text{BTPi}} = \underbrace{y_t^{\text{BTPi}} + \pi_t^{\text{swap}}}_{\text{synthetic nominal yield}} - y_t^{\text{nominal BTP}}$$

Why this differs from TIPS:

- **No ballooning risk.** Inflation revaluation paid out semiannually $\Rightarrow$ inflation factor resets to unity at each coupon $\Rightarrow$ no contamination from default exposure on accumulated revaluation.
- **Retail bondholder base.** Primary placement reserved to households + *premio fedeltà* for buy-and-hold $\Rightarrow$ preferred-habitat investors with no leverage, no margin calls, no forced unwinding.

Consequence: the BTP Italia basis behaves differently from TIPS during stress—and persists.

TIPS basis pre-2010 (Fleckenstein 2014):



TIPS basis post-2010: has vanished



Basis definition:

$$m_t^{\text{CDS–Bond}} = \text{CDS}_t^{\text{spread}} - z_t^{\text{spread}}$$

where z_t^{spread} is the bond's z-spread over the euro zero-coupon swap curve. A *negative* basis means credit protection via CDS is cheaper than the bond-implied credit spread.

The trade:

- Buy the corporate bond (earn z-spread); buy CDS protection (pay lower CDS premium).
- **Locked-in positive carry** until convergence or bond maturity.
- In default: bond recovers RR on par; CDS pays $100 - RR \Rightarrow$ **combined position returns par regardless of recovery** (provided CDS tenor $\geq$ bond residual life).

Universe: on-the-run iTraxx Europe Main 5Y constituents (investment-grade, liquid). All CDS centrally cleared (LCH/ICE) $\Rightarrow$ no counterparty credit risk on the protection leg.

Restriction to negative basis: the reverse trade requires reverse-repoing the corporate bond, subject to recall risk in stress (Bai and Collin-Dufresne, 2019).

Basis definition:

$$\text{Skew}_t = S_t^{\text{index, spread}} - \sum_{i=1}^N w_{i,t} S_{i,t}^{\text{single, spread}}$$

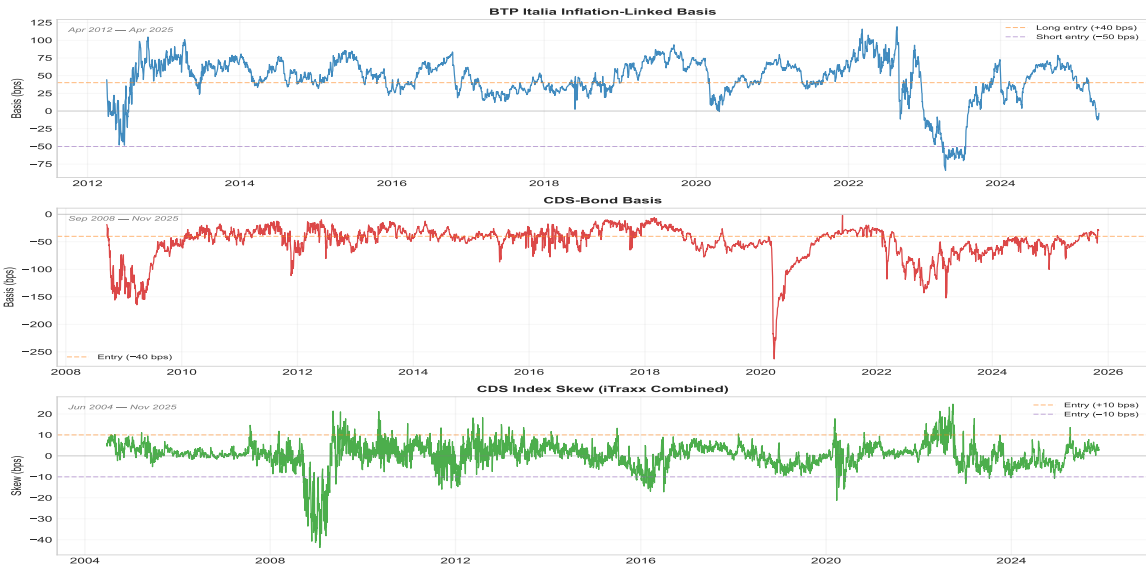
Quoted spreads in bps p.a.; $w_{i,t}$ are duration weights. A positive (negative) skew indicates that the index trades *wide* (*tight*) relative to the sum of its parts.

Why a non-zero skew can persist:

- **Liquidity differential.** Single-name CDS are far less liquid than the on-the-run index, with wider bid-ask spreads and shallower order books, especially in stress (Junge and Trolle, 2015). The skew is well documented as a market-wide liquidity proxy.
- **Strategic positioning by institutional investors.** Index CDS attract concentrated trading demand for macro credit hedging, making the index leg structurally tighter than the basket of single names that replicates it (Boyarchenko et al., 2016).
- **Execution complexity of the replicating basket.** Replicating the index up to ~ 125 single-name positions per sub-index versus a single index contract, with bid-ask costs and margin spread across many bilateral lines at trade inception and exit.

The trade: go long/short the index versus the duration-weighted single-name basket, depending on the sign of the skew. Entered on the on-the-run series and held to convergence (no rolling). Both legs centrally cleared.

Market-Level Basis Time Series



Following Duarte et al. (2007). Each individual trade is treated as a separate fictional fund (own entry/exit/return stream); the strategy index on any day is the weighted average of the returns of the funds active on that day.

Two weighting schemes:

Equal-Weighted (Duarte et al., 2007)

$$r_t^{\text{EW}} = \frac{1}{K_t} \sum_{k=1}^{K_t} r_{k,t}$$

Signal-Weighted (Rebonato and Ronzani, 2021)

$$r_t^{\text{SW}} = \sum_{k=1}^{K_t} \frac{|b_{k,0} - \theta_{t_k}|}{\sum_{j=1}^{K_t} |b_{j,0} - \theta_{t_j}|} r_{k,t}$$

$b_{k,0}$: basis at entry of trade k ; θ_{t_k} : expanding-window historical equilibrium up to entry. Weight $\propto$ distance from equilibrium $\rightarrow$ richer dislocation, larger allocation.

Implementation choices.

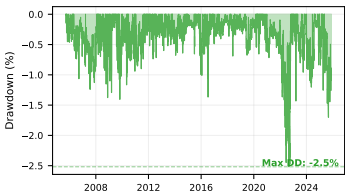
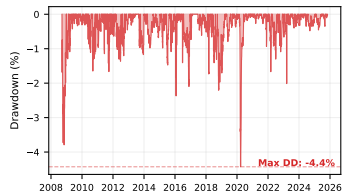
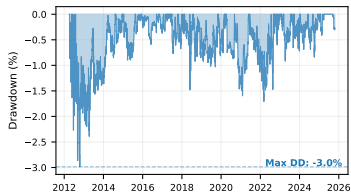
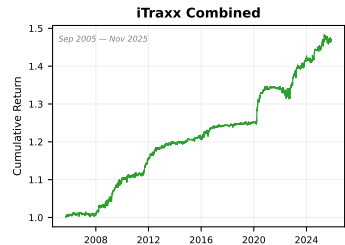
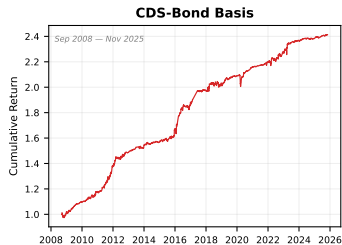
- **SW dominates EW** on Sharpe and MPPM for all three strategies $\rightarrow$ SW baseline; EW in appendix confirms robustness.
- Three-regime classification on iTraxx Main 5Y — *LOW* (< 60 bps), *MID* (60–100 bps), *HIGH* (> 100 bps) — drives **regime-dependent transaction costs** (entry/exit fees proportional to DV01).

Annualised, daily data, net of regime-dependent transaction costs.

	<i>BTP Italia</i>	<i>CDS–Bond Basis</i>	<i>iTraxx Combined</i>
Sample	Mar 2012 – May 2025	Sep 2008 – May 2025	Sep 2005 – May 2025
Mean return (%)	1.67	5.37	1.96
Volatility (%)	2.36	3.15	2.58
Sharpe ratio	0.71	1.71	0.76
Skewness	+0.60	+2.81	+0.43
Max drawdown (%)	3.0	4.4	2.5
Basis ρ_1 (daily)	0.98	0.94	0.87

- Sharpe ratios from **0.71** (BTP Italia) to **1.71** (CDS–Bond) — economically large.
- *Positive skewness* for all three (max +2.81 for the CDS–Bond basis) $\Rightarrow$ cuts against “picking up nickels in front of a steamroller”.
- Persistence $\rho_1 \in [0.87, 0.98]$ on the underlying basis $\Rightarrow$ slow-moving capital signature (Duffie, 2010).

Strategy Performance: Cumulative Returns and Drawdowns



MPPM and Volatility-Managed Performance

Manipulation-Proof Performance Measure (MPPM)

Strategy	MPPM (% p.a.)			Avg Return (% p.a.)	Vol Sharpe (% p.a.)	N
	$\rho = 2$	$\rho = 3$	$\rho = 4$			
CDS-Bond Basis	5.06	5.01	4.97	5.16	3.07	1.68 207
iTraxx <i>combined</i>	1.83	1.82	1.80	1.85	1.52	1.22 243
BTP-Italia	1.56	1.55	1.54	1.59	1.62	0.98 163

MPPM as in Goetzmann et al. (2007), based on monthly returns and reported in annualized percentage terms. All moments are annualized.

Volatility-Managed Portfolios (Moreira–Muir, 2017)

Strategy	Buy-and-Hold		Vol-Managed		Regression			N
	Ret.	Sharpe	Ret.	Sharpe	α	β	R^2	
BTP Italia	1.52	0.95	0.37	0.23	-0.22	0.39	0.150	158
CDS–Bond Basis	5.09	1.70	3.49	1.16	+1.68	0.36	0.127	202
iTraxx Combined	1.91	1.25	0.84	0.55	-0.16	0.53	0.279	242

- All three strategies deliver positive MPPM, with the highest utility-based performance for the CDS–Bond basis and iTraxx strategies.
- Volatility timing generates a sizable premium for the CDS–Bond basis, while effects are more modest for BTP Italia and iTraxx, consistent with their lower return variance.

We evaluate our three fixed-income arbitrage strategies under the established risk-factor models used in the literature:

- *Duarte, Longstaff & Yu (2007), Review of Financial Studies.*
Fixed-income arbitrage benchmark used to assess exposure to equity, credit and Treasury-based relative-value factors.
- *Fung & Hsieh (2004), Financial Analysts Journal.*
Trend-following and option-based factors widely used to benchmark hedge-fund strategies.
- *Brooks, Gould & Richardson (2020), Journal of Fixed Income.*
Active fixed-income premium decomposition: term, credit, EM, inflation-linked, and volatility premia.

US factors are mapped to their EUR counterparts. European equity factors use the Fama-French Europe datasets, converted from USD into EUR following the methodology of Glück, Hübel & Scholz (JPM, 2020). Trend-following factors (Fung-Hsieh) are global and directly applicable.

Duarte, Longstaff & Yu (2007) — 10-Factor FI Arbitrage Model

	α (% p.a.)	β_{Mkt-RF}	β_{SMB}	β_{HML}	β_{UMD}	β_{RS}	β_{RI}	β_{RB}	β_{R2}	β_{R5}	β_{R10}	R^2 adj	N
Panel A: US Factors													
<i>BTP Italia</i>	2.05*** (4.41)	-0.07*** (-3.40)	-0.03** (-2.19)	-0.04** (-2.44)	0.00 (0.07)	0.04*** (3.40)	0.04 (0.55)	-0.08 (-0.94)	0.09 (0.35)	-0.05 (-0.31)	0.04 (0.54)	0.132	161
<i>CDS Bond Basis</i>	5.13*** (6.66)	-0.01 (-0.43)	0.00 (0.08)	0.01 (0.40)	-0.03* (-1.75)	-0.01 (-0.39)	0.26*** (3.32)	-0.06 (-1.59)	0.11 (0.31)	-0.46* (-1.84)	0.12 (1.08)	0.174	204
<i>iTraxx Indices Skew</i>	1.81*** (4.64)	0.02 (1.55)	-0.02* (-1.70)	-0.02 (-1.28)	-0.02* (-1.91)	-0.01 (-1.44)	-0.02 (-0.37)	-0.02 (-0.71)	0.34 (1.53)	-0.14 (-0.99)	0.05 (0.84)	0.051	240
Panel B: EUR Factors													
<i>BTP Italia</i>	1.65*** (3.36)	-0.05** (-1.99)	-0.06*** (-2.97)	-0.05** (-2.05)	0.02 (0.99)	0.02 (1.27)	-0.01 (-0.08)	0.03 (0.36)	0.05 (0.11)	-0.03 (-0.14)	-0.03 (-0.27)	0.108	162
<i>CDS Bond Basis</i>	5.14*** (5.59)	0.00 (0.08)	0.02 (0.53)	0.04 (1.13)	-0.03 (-1.38)	-0.03** (-2.01)	0.47*** (3.87)	-0.07* (-1.70)	-0.53 (-0.80)	0.15 (0.32)	-0.17 (-1.05)	0.144	205
<i>iTraxx Indices Skew</i>	2.26*** (4.64)	-0.02 (-1.20)	-0.04*** (-2.60)	-0.07*** (-2.58)	-0.03** (-2.27)	0.01 (1.16)	0.06 (1.00)	-0.02 (-0.99)	0.02 (0.05)	0.06 (0.26)	-0.07 (-0.90)	0.091	241

Fung & Hsieh (2004) — 7-Factor Hedge Fund Model

	α (%)	$\beta_{S\&P}$	β_{SC-LC}	β_{BdOpt}	β_{FXOpt}	β_{ComOpt}	β_{10Y}	$\beta_{CredSpr}$	R^2 adj	N
Panel A: US Factors										
<i>BTP Italia</i>	1.80*** (3.64)	-0.02* (-1.96)	-0.02** (-2.08)	0.12 (0.72)	-0.01 (-0.03)	0.33 (1.05)	0.00 (0.01)	0.28 (1.18)	0.104	162
<i>CDS Bond Basis</i>	5.28*** (6.68)	-0.02 (-1.58)	0.01 (0.53)	0.09 (0.35)	-0.47 (-1.49)	-0.09 (-0.27)	-0.98*** (-3.98)	-1.73*** (-5.35)	0.185	205
<i>iTraxx Indices Skew</i>	1.69*** (4.32)	0.02 (1.46)	-0.02** (-2.09)	0.10 (0.48)	0.17 (0.89)	0.13 (0.48)	-0.07 (-0.57)	0.16 (0.78)	0.029	241
Panel B: EUR Factors										
<i>BTP Italia</i>	1.86*** (3.64)	-0.04*** (-2.99)	-0.06** (-2.40)	0.34* (1.80)	-0.08 (-0.39)	0.34 (0.92)	0.28 (1.44)	-0.53 (-1.13)	0.111	162
<i>CDS Bond Basis</i>	4.67*** (5.38)	-0.03 (-1.60)	0.04 (1.22)	0.21 (0.88)	-0.71* (-1.80)	-0.59 (-1.64)	-0.88** (-2.38)	-0.74 (-1.32)	0.112	195
<i>iTraxx Indices Skew</i>	3.48*** (6.05)	-0.00 (-0.28)	-0.07** (-2.02)	0.12 (0.53)	-0.13 (-0.56)	0.59* (1.83)	-0.34 (-1.37)	-0.12 (-0.38)	0.051	195

Brooks, Gould & Richardson (2020) — Active FI Premia

	α (% p.a.)	β_{Term}	$\beta_{GlobalTerm}$	$\beta_{GlobalAgg}$	$\beta_{Infl.Linkers}$	$\beta_{CorpCredit}$	$\beta_{EmergDebt}$	$\beta_{EmergCurr}$	β_{USTVol}	R^2 adj	N
Panel A: US Factors											
<i>BTP Italia</i>	1.88*** (3.65)	0.15 (0.98)	-0.27 (-0.88)	0.13 (0.31)	-0.02 (-0.82)	-0.14** (-2.37)	0.09* (1.72)	-0.02 (-0.91)	0.00 (0.15)	0.115	163
<i>CDS Bond Basis</i>	4.61*** (5.38)	0.01 (0.08)	-0.20 (-0.43)	0.42 (0.73)	-0.03 (-0.44)	-0.00 (-0.02)	0.07 (0.97)	0.02 (0.43)	0.01 (0.40)	0.104	207
<i>iTraxx Indices Skew</i>	1.57*** (4.36)	-0.05 (-0.45)	-0.10 (-0.43)	0.21 (0.68)	-0.01 (-0.17)	0.04 (1.08)	-0.07* (-1.82)	0.03 (1.49)	0.04** (2.00)	0.098	243
Panel B: EUR Factors											
<i>BTP Italia</i>	1.96*** (3.28)	-0.07 (-1.47)	-0.36 (-1.43)	0.51* (1.87)	-0.07** (-2.01)	-0.06 (-1.23)	-0.03 (-0.93)	0.03 (1.01)	0.00 (0.21)	0.009	163
<i>CDS Bond Basis</i>	5.07*** (5.87)	-0.02 (-0.17)	-0.05 (-0.12)	0.25 (0.70)	-0.02 (-0.24)	-0.08 (-1.52)	0.15** (2.21)	-0.02 (-0.31)	-0.02 (-1.30)	0.039	207
<i>iTraxx Indices Skew</i>	1.96*** (4.73)	0.03 (0.32)	-0.24 (-1.09)	0.26 (1.34)	-0.02 (-0.50)	0.01 (0.81)	-0.03 (-1.30)	-0.02 (-1.16)	-0.02* (-1.76)	0.008	243

Cross-Model Synthesis: Alpha is Robust and *State-Dependent*

Full-sample alpha (% p.a.)

	<i>BTP Italia</i>	<i>CDS–Bond</i>	<i>iTraxx</i>
Duarte et al. (2007)	+1.65***	+5.14***	+2.26***
Brooks et al. (2020)	+1.93***	+5.07***	+1.96***
Fung & Hsieh (2004)	+1.86***	+4.67***	+3.48***

Conditional Alpha (*Ferson–Schadt*, α_1 per 1σ stress)

	<i>BTP Italia</i>	<i>CDS–Bond</i>	<i>iTraxx</i>
Duarte et al. (2007)	+1.09*	+2.87***	+1.06***
Brooks et al. (2020)	+1.50**	+2.90***	+1.26***
Fung & Hsieh (2004)	+1.39*	+3.31***	+2.81***

Regime decomposition (*HIGH: iTraxx Main 5Y > 100 bps*)

	Duarte et al. (2007)		Brooks et al. (2020)		Fung & Hsieh (2004)	
	NORM	HIGH	NORM	HIGH	NORM	HIGH
<i>BTP Italia</i>	+1.65***	+4.74**	+1.42***	+4.13**	+1.46***	+5.62***
<i>CDS–Bond Basis</i>	+3.77***	+8.63***	+3.63***	+9.69***	+2.95***	+9.56***
<i>iTraxx Combined</i>	+1.98***	+2.71***	+1.38***	+4.38***	+2.61***	+7.25***

- *Full-sample*: alpha consistent across the three frameworks.
- *Regime decomposition*: HIGH-stress alpha is **2–4×** NORMAL alpha—in every cell.
- *Conditional alpha*: $\alpha_1 > 0$ everywhere; sig. at 1% for the institutional strategies $\Rightarrow$ **slow-moving capital signature**.

The three benchmarks of Section 3:

- Alpha positive and significant across all 9 cells
- $\bar{R}^2 \in [1\%, 18\%]$ — bulk of return variation *unexplained*

Two open questions:

- ① Are there *other* risk factors — outside the standard benchmarks — that span these returns?
- ② If alpha survives even with a much richer factor universe, the case for genuine mispricing strengthens.

Strategy:

- Assemble **85 candidate factors** from the empirical asset pricing, intermediary, and macro-finance literatures.
- Two complementary approaches to exploit the broader factor universe: **rolling PCA** (Ludvigson and Ng, 2009) and **Adaptive Elastic Net** with stability selection (Zou and Zhang, 2009; Meinshausen and Bühlmann, 2010).

Factor universe (organized in 6 families)

- *Credit risk* (e.g., CDS indices, excess bond premium, sovereign spreads)
- *Liquidity & funding stress* (e.g., OIS spreads, repo indicators, noise/illiquidity measures)
- *Equity / intermediary proxies* (e.g., BAB, value, momentum)
- *Volatility* (e.g., VIX/V2X, MOVE, option-implied measures, trend-following)
- *Macro uncertainty & policy* (e.g., uncertainty indices, EPU, financial conditions)
- *Rates & term structure* (e.g., term, slope, swap spreads)

$2^{85} \approx 3.87 \times 10^{25}$ subsets $\Rightarrow$ exhaustive search infeasible $\Rightarrow$ penalised methods needed.

Setup (Ludvigson and Ng, 2009):

- Rolling PCA, window $W = 24$ months, $K = 8$ components. Across rolling windows, the eight components explain 80.8% of variance on average (75.7%–87.8%).
- Standardisation *within* each estimation window $\Rightarrow$ no look-ahead.
- Both contemporaneous ($PC_t \rightarrow r_t$) and predictive ($PC_t \rightarrow r_{t+1}$) timing.

Strategy	Contemporaneous		Predictive	
	α (% p.a.)	$\bar{R}^2$	α (% p.a.)	$\bar{R}^2$
<i>BTP Italia</i>	+1.61*** (3.48)	0.054	+1.68*** (4.05)	−0.005
<i>CDS–Bond Basis</i>	+5.87*** (5.74)	0.120	+5.43*** (5.91)	0.050
<i>iTraxx Combined</i>	+2.21*** (5.01)	0.049	+2.07*** (5.38)	0.041

- Alpha survives at 1%** for all three strategies, both timings.
- $\bar{R}^2$ stays low (0–12%) $\Rightarrow$ we turn to **machine-learning** methods (AEN) to identify factors that better span these returns.

Two-stage penalised regression of Zou and Zhang (2009):

$$\hat{\beta}^{\text{AEN}} = \arg \min_{\beta} \left\{ \frac{1}{T} \sum_t (r_t - \alpha - \beta' f_t)^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \sum_{j=1}^p w_j |\beta_j| \right\}, \quad w_j = |\hat{\beta}_j^{\text{EN}}|^{-\gamma}, \quad \gamma = 1.$$

- **Adaptive weights** (Stage 1 EN $\rightarrow$ Stage 2 weights): factors with large Stage-1 coefficients are penalised *less*; near-zero ones are penalised *more* $\Rightarrow$ restores the *oracle property* (LASSO is inconsistent under correlated factors).
- ℓ_2 **ridge** $\Rightarrow$ *grouping effect*: highly correlated factors selected/dropped *together*, not arbitrarily.
- **Tuning** (λ_1, λ_2) via Hannan–Quinn (HQC); coordinate descent (Friedman et al., 2010).
- **Stability selection** (Meinshausen and Bühlmann, 2010): $B = 100$ stationary bootstrap (expected block 9m), retain factor only if selection frequency $\hat{\pi}_j \geq 80\%$ $\Rightarrow$ formal guarantee on expected false selections.
- **Post-selection OLS** on stable set, with Newey–West HAC $\Rightarrow$ unbiased inference + sparsity.

Variable	Coefficient	t-stat	p-value
α	+1.39***	3.34	0.00
SMB _{EU}	-0.04**	-2.30	0.02
ILLIQ	+0.64**	2.50	0.01
SS10Y	-0.02***	-3.26	0.00
UMD _{EU}	+0.03**	1.96	0.05
CDX _{IG}	+0.01	1.60	0.11
LIBOR _{OIS}	+0.49*	1.90	0.06

Model fit

T (obs)	157
k (factors)	6
R^2	0.2474
R^2_{adj}	0.2173
Durbin–Watson	2.0351

(α) annualized %.
 *** $p < 1\%$, ** $p < 5\%$,
 * $p < 10\%$.

- SMB_{EU}: European equity size factor (small – big caps).
- ILLIQ: Change in the Amihud illiquidity shock measure.
- SS10Y: 10Y EUR swap spread (par swap rate – 10Y Bund yield).
- UMD_{EU}: European equity momentum factor (winners – losers).
- CDX_{IG}: First difference of the CDX Investment Grade index CDS spread.
- LIBOR_{OIS}: Libor–OIS spread (3M LIBOR – 3M OIS).

Fama and French (2017)
Amihud and Mendelson (2015)
Collin-Dufresne et al. (2001)
Carhart (1997)
Klaus and Rzepkowski (2009)
Nyborg and Ostberg (2014)

AEN Stable-Factor Model: CDS–Bond Basis

Variable	Coefficient	t-stat	p-value
α	+3.94***	3.66	0.00
ΔUF	-4.27***	-2.61	0.01
RI_{EU}	+0.16***	3.27	0.00
$PB_{EU_CDS_1Y}$	+0.01**	2.02	0.04
$EMERG_{FX}$	+0.06**	1.99	0.05
$SS5Y$	+0.02**	2.33	0.02
$CRED_{SPR_US}$	-0.51**	-2.20	0.03
EP_{SVIX_1M}	+3.25***	2.82	0.00
$PTFSFX$	-0.45*	-1.74	0.08

Model fit	
T (obs)	197
k (factors)	8
R^2	0.2969
R^2_{adj}	0.2670
Durbin–Watson	2.0233

(α) annualized %.
 *** $p < 1\%$, ** $p < 5\%$,
 * $p < 10\%$.

- ΔUF : Change in financial uncertainty index.
- RI_{EU} : Euro investment-grade industrial bond return factor.
- $PB_{EU_CDS_1Y}$: European prime broker 1Y CDS spread (counterparty risk).
- $EMERG_{FX}$: Equal-weighted basket of emerging market currencies vs USD.
- $SS5Y$: 5Y EUR swap spread (par swap rate – 5Y Bobl yield).
- $CRED_{SPR_US}$: US credit spread factor (Moody's BAA – AAA yield).
- EP_{SVIX_1M} : Option-implied equity premium via SVIX, 1-month horizon.
- $PTFSFX$: Currency trend-following factor (lookback straddles on FX futures).

Ludvigson et al. (2021)
Collin-Dufresne et al. (2001)
Klaus and Rzepkowski (2009)
Brooks et al. (2020)
Collin-Dufresne et al. (2001)
Fama and French (1989)
Martin (2017)
Fung and Hsieh (2004)

AEN Stable-Factor Model: iTraxx Combined

Variable	Coefficient	t-stat	p-value
α	+0.47	1.04	0.30
EP_{SVIX_1M}	+3.31***	2.89	0.00
$LIBOR_{OIS}$	+0.48**	2.21	0.03
$\Delta V2X$	-0.02**	-2.55	0.01
ΔUR	+3.01**	2.35	0.02
$\Delta FAILS_{PCT_TSY}$	+0.40***	3.20	0.00

Model fit

T (obs)	237
k (factors)	5
R^2	0.2521
R^2_{adj}	0.2359
Durbin–Watson	2.2489

(α) annualized %.
 *** $p < 1\%$, ** $p < 5\%$,
 * $p < 10\%$.

- EP_{SVIX_1M} : Option-implied equity premium via SVIX, 1-month horizon. *Martin (2017)*
- $LIBOR_{OIS}$: Libor–OIS spread (3M LIBOR – 3M OIS). *Nyborg and Ostberg (2014)*
- $\Delta V2X$: Change in V2X (30-day implied vol on Euro STOXX 50). *Chung et al. (2019)*
- ΔUR : Change in real-activity uncertainty index. *Ludvigson et al. (2021)*
- $\Delta FAILS_{PCT_TSY}$: Change in U.S. Treasury fails-to-deliver / total debt outstanding. *Fleckenstein et al. (2014)*

Cross-Method Comparison: Benchmarks vs PCA vs AEN

Method	<i>BTP Italia</i>		<i>CDS–Bond</i>		<i>iTraxx</i>	
	α (%)	$\bar{R}^2$	α (%)	$\bar{R}^2$	α (%)	$\bar{R}^2$
Duarte (k=10)	+1.65***	0.108	+5.14***	0.144	+2.26***	0.091
Brooks (k=8)	+1.96***	0.009	+5.07***	0.039	+1.96***	0.008
Fung–Hsieh (k=7)	+1.86***	0.111	+4.67***	0.112	+3.48***	0.051
PCA (K=8)	+1.61***	0.054	+5.87***	0.120	+2.21***	0.049
AEN (k=6/8/5)	+1.39***	0.217	+3.94***	0.267	+0.47	0.236

- $\bar{R}^2$ jumps from 1–18% (benchmarks) to ~ 22 –27% (AEN): the AEN-selected factors absorb a much larger share of strategy return variation than any classical benchmark.
- Alpha survives PCA and AEN for BTP Italia and CDS–Bond Basis: a residual premium remains. For iTraxx Combined alpha collapses ($t \approx 1.04$): the skew premium is fully absorbed by AEN factors.

The AEN-selected factor sets are **economically interpretable** and **differ sharply across strategies**, mapping onto the economic structure of each trade.

- **BTP Italia** → *liquidity, funding, term-structure, and equity style* (Amihud illiquidity, Libor–OIS, 10Y EUR swap spread, SMB).
- **CDS–Bond Basis** → *the classical intermediary asset-pricing channel* (European PB CDS, Δ UF, EP_SVIX, 5Y swap spread) — exactly the channel slow-moving capital predicts for a trade requiring continuous repo financing of the cash bond (Boyarchenko et al., 2016).
- **iTraxx Combined** → *aggregate funding stress and market-wide volatility* (EP_SVIX, Libor–OIS, Δ V2X, Treasury fails) — consistent with the characterisation of the index skew as a market-wide liquidity proxy (Junge and Trolle, 2015). The collapse of residual alpha under these factors reinforces this reading.

What we have established:

- Alpha exists, robust to benchmarks, PCA on 85 factors, AEN with stability selection.
- Alpha rises in stress (regime decomposition, Ferson–Schadt, rolling).

The deeper question:

Why does alpha persist, and are the three mispricings linked?

Theoretical anchors:

- Duffie (2010) — qualitative framework of slow-moving capital.
- Brunnermeier and Pedersen (2009) — formal model with Proposition 6 on commonality of liquidity.
- Mitchell and Pulvino (2012) — empirical cross-strategy mechanism (*several months* timescale).
- Boyarchenko et al. (2016) — post-crisis equilibrium and funding-dependence.
- Fleckenstein et al. (2014) — cross-arbitrage spanning methodology.

Five Testable Predictions of Slow-Moving Capital

Prediction	
P1	Mispricings serviced by a common pool of intermediary capital should <i>co-move</i> .
P2	The co-movement should <i>intensify</i> when intermediary capital is scarce.
P3	Mispricing changes in one strategy should <i>predict</i> those of others, propagating asymmetrically along the liquidity gradient.
P4	The <i>persistence</i> of mispricing levels should reflect the funding-dependence of each arbitrage trade, with strategies that require continuous balance-sheet funding displaying the longest half-lives.
P5	A common factor extracted from the co-movement should be explained by intermediary balance-sheet variables, <i>not</i> macro fundamentals.

Arbitrageur (all three strategies): Institutional —
hedge funds, dealer prop desks, leveraged accounts.

Institutional bondholder base
CDS–Bond Basis, iTraxx Combined
Counterparty subject to margin calls,
VaR limits, balance-sheet constraints.

VS

Retail bondholder base
BTP Italia
Households (loyalty bonus, inflation
hedge for IT investors). No lever-
age, no margin calls, buy-and-hold.

Why this matters (Brunnermeier and Pedersen, 2009 + Vayanos and Vila, 2021):

- **Institutional bondholder pair:** stress → VaR spikes → risk budget exhausted → *both* bondholders + arbitrageurs unwind.
- **Retail bondholder:** insulated from funding shocks → no forced selling → basis remains open for arbitrageurs who can still deploy capital.

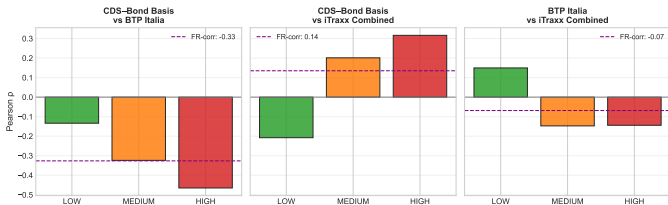
Identification: *if co-movement were driven by macro, all three would move in the same direction regardless of who holds the bond.*

Pearson correlations of mispricing changes $\Delta m_{i,t}$, unconditional. HAC inference.

Pair	Pearson ρ	t-HAC	p-value	N
CDS–Bond Basis vs BTP Italia	−0.298 * **	−3.01	0.003	206
CDS–Bond Basis vs iTraxx Combined	+0.191*	1.70	0.089	206
BTP Italia vs iTraxx Combined	−0.100	−1.07	0.285	156

- **Within institutional pair** (CDS–Bond $\leftrightarrow$ iTraxx): positive correlation $\Rightarrow$ consistent with a common pool of dealer capital.
- **Across investor bases** (BTP Italia pairs): negative correlation $\Rightarrow$ first signal of segmentation.
- Foundational anchor: Duffie (2010) catastrophe-insurance analogy + Brunnermeier and Pedersen (2009) commonality of liquidity (Prop. 6(i)).

Regime correlations (Fig. 14):



Interaction $\hat{\beta}_1$:

$$\Delta m_i = \beta_0 \Delta m_j + \beta_1 (\Delta m_j \times z_t) + \gamma z_t + \varepsilon$$

Direction	$\hat{\beta}_1$
BTP $\rightarrow$ CDS-Bond Basis	-0.0866
CDS-Bond Basis $\rightarrow$ BTP	-0.6400 * **
iTraxx $\rightarrow$ CDS-Bond Basis	+0.0326
CDS-Bond Basis $\rightarrow$ iTraxx	+0.0799*
iTraxx $\rightarrow$ BTP	-0.6186
BTP $\rightarrow$ iTraxx	-0.0197

- Institutional pair (CDS-Bond $\leftrightarrow$ iTraxx):** ρ rises monotonically LOW $\rightarrow$ HIGH $\Rightarrow$ *commonality of fragility* (Brunnermeier and Pedersen, 2009, Prop. 6(ii)).
- Cross-base pairs (BTP $\leftrightarrow$ inst.):** deeper negative correlation in HIGH (with sign reversal for BTP $\leftrightarrow$ iTraxx) $\Rightarrow$ *flight to quality* (Brunnermeier and Pedersen, 2009, Prop. 6(iv)).
- Interaction β_1 confirms continuous stress amplification with sign asymmetry.

$$\Delta m_{i,t} = \alpha + \sum_{j \neq i} \sum_{l=0}^2 \beta_{j,l} \Delta m_{j,t-l} + \varepsilon_t \quad \text{— design of Fleckenstein et al. (2014).}$$

Cross-strategy spanning ($\bar{R}^2$):

Dependent: $\Delta m_{i,t}$	$\bar{R}^2$
CDS–Bond Basis	9.1%
BTP Italia	10.0%
iTraxx Combined	1.0%

Granger causality (VAR on Δm):

From	To	p -value
BTP	→ CDS–Bond Basis	0.818
iTraxx	→ CDS–Bond Basis	0.011**
CDS–Bond Basis	→ BTP	0.080*
iTraxx	→ BTP	0.007***
CDS–Bond Basis	→ iTraxx	0.904
BTP	→ iTraxx	0.490

- **Hierarchy:** iTraxx (most liquid) → CDS–Bond → BTP Italia.
- BTP Italia is the **most predictable** ($\bar{R}^2$ highest): retail-held basis adjusts *last*, with 0–1 months delay.
- Consistent with funding-liquidity feedback of Brunnermeier and Pedersen (2009) and cross-strategy capital reallocation of Mitchell and Pulvino (2012).

P4: Persistence Reflects Funding-Dependence

$m_{i,t} = c_i + \phi_i m_{i,t-1} + u_{i,t}$, *monthly mispricing levels*. Half-life $h^* = \ln(0.5) / \ln(\phi_i)$.

Strategy	ϕ	HAC t	Half-life (months)
CDS–Bond Basis	0.791	(14.5)	3.0
BTP Italia	0.727	(13.9)	2.2
iTraxx Combined	0.726	(10.0)	2.2

- **CDS–Bond is the most persistent** (half-life ≈ 3 months): the only strategy that requires *continuous repo financing* of the cash bond on dealer balance sheets (Boyarchenko et al., 2016).
- **BTP Italia and iTraxx mean-revert faster and are essentially identical** (half-life ≈ 2 months at $\phi \approx 0.73$): the two less funding-intensive strategies are statistically indistinguishable from each other in monthly persistence.
- Persistence is segregated by **funding-dependence of execution**: only the strategy requiring continuous repo financing of a cash bond stands apart from the two derivative-based bases.

P5: Common Factor → Intermediary Capital, *not* Macro

Factor	Coefficient	p -value
Panel A: Core Intermediary Proxies		$T = 149, \bar{R}^2 = 0.231$
HKM_IC	-4.292 * *	0.026
LIBOR_OIS	+3.336 * **	0.002
PB_EU_CDS_5Y	-0.018 * *	0.049
ILLIQ	+1.730 * **	0.005
Panel C: Macroeconomic Placebo		$T = 149, \bar{R}^2 = 0.112$
Δ UM	+0.234	0.984
Δ UR	+11.027	0.194
5Y5Y_INFL	-0.579	0.549
EPU_EU	+0.003	0.203

- **Panel A:** all four intermediary proxies significant with the predicted sign $\Rightarrow$ direct empirical content of Duffie (2010) and Prop. 6(i) of Brunnermeier and Pedersen (2009).
- **Panel C (placebo):** *none* of the four macro proxies is significant; model not jointly significant (F -test $p=0.23$) $\Rightarrow$ confirms common factor is balance-sheet, not macro.

All Five Predictions Find Support

P	Statement	Empirical evidence	
P1	Cross-market correlation	$\rho_{\Delta m} = +0.19^*$ (CDS–Bond Basis vs iTraxx Combined)	✓
P2	Stress amplification	$\rho_{\text{LOW}} = -0.21 \rightarrow \rho_{\text{HIGH}} = +0.32$	✓
P3	Lead-lag transmission	BTP Italia $\bar{R}^2 = 10.0\%$ (iTraxx leads at 1%)	✓
P4	Persistence reflects funding-dependence	CDS–Bond Basis $\sim 3.0\text{m}$; BTP Italia, iTraxx Combined $\sim 2.2\text{m}$ (essentially identical)	(partial)
P5	Common factor = intermediary, not macro	Intermediary $\bar{R}^2 = 23.1\%$ vs Macro placebo $\bar{R}^2 = 11.2\%$ (n.s.)	✓

Three reinforcing mechanisms:

- **Regulatory cost of intermediary capacity** (Boyarchenko et al., 2016):
post-crisis SLR and Basel III have permanently raised the cost of dealer balance sheet, lifting the equilibrium level of basis trades — evident in the high persistence of CDS–Bond ($\phi = 0.79$, P4).
- **Slow-moving capital across strategies** (Duffie, 2010; Mitchell and Pulvino, 2012):
capital reallocates across the dealer network on a horizon of several months. iTraxx adjusts first, BTP Italia last (P3), and the common factor is intermediary balance-sheet, not macro fundamentals (P5).
- **Investor-base segmentation as a structural feature:**
BTP Italia retail bondholders generate a preferred-habitat (Vayanos and Vila, 2021) structurally insulated from institutional unwinding — the basis remains open for arbitrageurs who can deploy capital, but the funding-dependence of execution determines who can.

Bottom line: the geography of slow-moving capital is determined not by asset-class labels but by the network of intermediaries that service these markets — and ultimately by who holds the bonds.